Curriculum Vitae

PERSONAL DATA

Surname: Eyrea Irazú

Name: María Emma

Place of birth: La Plata, Buenos Aires, Argentina

Address: Departamento de Matemática, Universidad Nacional de La Plata

E-Mail address: maemma@mate.unlp.edu.ar

Professional Profile: I am María Emma Eyrea Irazú, a Mathematician with a Bachelor's and a Ph.D. from the National University of La Plata (UNLP). My academic and professional career revolves around research and teaching in applied mathematics, with a particular focus on geometric mechanics and dynamical systems with magnetic terms. I have held various teaching positions at the Faculty of Exact Sciences at UNLP, including roles as a Senior Teaching Assistant and Adjunct Professor in multiple courses.

Throughout my career, I have been awarded doctoral and postdoctoral fellowships by CONICET, allowing me to deepen my research and contribute to the scientific community. I have actively participated in numerous national and international conferences, presenting my work on geometry, mechanics, and mathematical physics. My research has been published in high-impact scientific journals, and I have had the opportunity to conduct research stays at renowned institutions such as the Institute of Mathematical Sciences (ICMAT) in Spain.

Beyond research, I am passionate about education and have been involved in secondary and tertiary teaching, leading outreach projects to improve mathematics education at various levels. My dedication to both research and teaching drives me to continue exploring new challenges and contributing to the academic and scientific community.

Educational and professional experience:

Education:

- 2019, December 18th – **PhD in Mathematics**, Universidad Nacional de La Plata, Argentina (UNLP), Advisors: Marcela Zuccalli and Leonardo Jesús Colombo. Thesis title: Geometric and numerical aspects of mechanical systems with magnetic terms. Mark: Sobresaliente (outstanding).
- 2013, August – **Tertiary Course on pedagogical training for higher-school level**, Instituto Superior del Profesorado Juan N. Terrero, La Plata, Argentina. Mark: 9.65 Sobresaliente (outstanding).
- 2004-2012, March – **Licenciatura en Matemática** (equivalent to Bachelor's Degree in Mathematics), Universidad Nacional de la Plata, Argentina. Mark 8 out of 10.
- 2015-present – **English Courses**, Intermediate Level, Speaker Institute, La Plata, Argentina.

- 2014 – 2016 – Postgraduate Courses: Introduction to financial calculation, Symmetries in Mechanics, Introduction to Scientific knowledge methods and Introduction of Geometric Theory of PDE´s. Universidad Nacional de La Plata, Argentina (UNLP). Marks: 10 Sobresaliente (average).

Professional experience:

- 04/2020 – 06/2023 – CONICET Postdoctoral Fellowship. Argentina.
- 04/2015 – 03/2020 – CONICET Doctoral Fellowship. Argentina.
- 09/2023 – Present – Professor, FCE, UNLP. Argentina.
- 05/2024 – Present – Head of Practical Assignment and researcher at CMaLP, FCE, UNLP. Argentina.
- 02/2026 – Present – Subject Coordinator, FCE, UNLP. Argentina.
- 04/2023 – 05/2024 – Assistant professor and researcher at CMaLP, FCE, UNLP. Argentina.
- 04/2015 – 08/2023 – Head of Practical Assignment, FCE, UNLP. Argentina.
- 07/2014 – 03/2022 – Instructor Precalculus of Leveling for new students at the University, UNLP. Argentina.
- 07/2013 – 04/2015 – Teaching Assistant, FCE, UNLP. Argentina.
- 05/2008 – 02/2024 – Secondary School Math Teacher in different levels. DGCYE. Argentina.
- V category of Teacher-researcher from National University of La Plata. Argentina.
- DI4 category of Teacher-researcher from National University of La Plata. Argentina.

Teaching experience: 15 years of experience as secondary level teaching in different schools and different levels, as Math Teacher in Buenos Aires, Argentina. 13 years of experience as Instructor teacher for new student at the computing, engineering, exact sciences in courses of: Mathematics 0, Mathematics for engineering. 2 year and 6 months of experience as teaching assistant in the department of mathematics, Science Faculty, Computer Science Faculty and Department of Basics Sciences, Engineering Faculty (Universidad Nacional de La Plata), Argentina in classes of: Mathematics I, Mathematics for engineering and Mathematics II. 11 years of experience as Head of Practical Assignment (Responsible of Problems sessions, midterm exams and coordination of grade) in the department of mathematics, Science Faculty and Computer Science Faculty (Universidad Nacional de La Plata), Argentina in classes of: Mathematics I, Mathematics II, Mathematical Analysis I, Mathematical Analysis II and Differential Geometry. 2 year of experience as Professor in the department of mathematics, Science Faculty (Universidad Nacional de La Plata), Argentina in classes of: Algebra and Mathematical Analysis II.

Publications.

Published or accepted for publication papers: The information on citations was obtained via Google Scholar and excludes self-citations. The relevance and impact of my publications is explained in Sec. Main contributions to the field below.

Journal papers:

1. Symmetries and periodic orbits in simple hybrid Routhian systems. **Nonlinear Analysis: Hybrid systems**. Vol. 36, 100857, (2020). (With Leonardo Colombo).
2. A note on Hybrid Routh reduction for time dependent Lagrangian systems**. Journal of Geometric Mechanics**. Vol. 12, No. 2, pp. 309-321, (2020). (With Leonardo Colombo, Eduardo García-Toraño).
3. Hamilton-Jacobi theory for nonholonomic and forced hybrid mechanical systems. **Geometric Mechanics**, Vol. 01, No. 02, pp. 123-152 (2024). (With Leonardo Colombo, Manuel de Leon, Asier López Gordón).
4. Generalized hybrid momentum maps and reduction by symmetries of forced mechanical systems with inelastic collisions. **Journal of Mathematical Physics**. Vol. 66, No. 6, 062901 (2025). (With Leonardo Colombo, Manuel de León, Asier López Gordón).

Conference (refereed) proceedings:

1. Hybrid Routhian reduction for simple hybrid forced Lagrangian systems. **Proceedings of European Control Conference** (ECC), **2021**, pp. 345-350. (With Leonardo Colombo, Manuel de León, Asier López Gordón).
2. Reduction by symmetries of simple hybrid mechanical systems. **Proceedings of 7th IFAC Workshop on Lagrangian and Hamiltonian Methods in Nonlinear Control, 2021**, 54 (19), pp. 94-99. (With Anthony Bloch, Leonardo Colombo).
3. Reduction by symmetries and higher-order magnetic systems. **Proceedings of VII MACI**, **2021**, pp. 163-166. (With Leonardo Colombo, Marcela Zuccalli).
4. Lagrangian mechanics and Routh reduction for an ASV-UAV system based on cluster-space formulation. **Proceedings of VIII MACI**, **2023**, 127-130. (With Leonardo Colombo, Maria Daniela Sanchez, Marcela Zuccalli).
5. Time-dependent Lagrangian systems on Lie groups. **Proceedings of IX MACI**, **2025**, 238-241. (With Leonardo Colombo, Maria Daniela Sanchez, Marcela Zuccalli).
6. Reduction of hybrid Hamiltonian systems with non-equivariant momentum maps. **Proceedings of Geometric science of information**, (GSI) **2025**. LNCS 16035, pp. 303-310. (With Leonardo Colombo, Maria Eugenia García Asier López Gordón, Marcela Zuccalli).
7. Homogeneous Bi-Hamiltonian structures and integrable contact systems. **Proceedings of Geometric science of information**, (GSI) **2025**. LNCS 16035, pp. 30-39. (With Leonardo Colombo, Manuel de León, Asier López Gordón).
8. On time-dependent Lagrangian systems over Lie groups and geometric integrators. **Proceedings of XLI Argentine Congress of Computational Mechanics** (MECOM), **2025**. Vol XLII, pp. 715-725. (With Leonardo Colombo, David Martín de Diego, Maria Daniela Sanchéz, Marcela Zuccalli).

Invited presentations to peer-reviewed, internationally established conferences and/or international advanced schools.

I have communicated the developments in my research in leading international conferences in the interface between applied mathematics, engineering and theoretical physics as an invited speaker.

- LINAR meeting group on robotics and mechanics. UDESA, San Fernando (Buenos Aires), Argentina, April, 2022.
- ICMAT meeting group on Geometry, Mechanics and Control, Madrid, Spain, March, 2019.

Contributed presentations/posters to peer-reviewed, internationally established conferences, international advanced schools and/or lecture at seminars.

Contributed talks and posters at conferences:

- XLI Argentine Congress of Computational Mechanics (MECOM). Buenos Aires, Argentina, November 11-14, 2025. (talk)
- Congress of the Americas (MCA). Miami, United States, July 21-25, 2025. (talk)
- Inspiring Mathematics. Chicago, United States, July 14-18, 2025. (talk)
- I Interdisciplinary Meeting: Physics and Mathematics. La Plata, Argentina, November 14-15, 2024. (talk)
- IV Argentine meeting of Geometric Mechanics and Mathematical Physics. Bahia Blanca, Argentina, April 29 to May 3, 2024. (poster)
- Meeting of Wildlife Monitoring Bots. La Plata, Argentina, March 11-12, 2024. (talk)
- Congress of Argentine Mathematical Union. Salta, Argentina, September 19-22, 2023. (talk)
- III Argentine meeting of Geometric Mechanics and Mathematical Physics. La Plata, Argentina, August, 24-26, 2022. (poster)
- VII Iberoamerican meeting on Geometry, Mechanics and Control. Bahia Blanca, Argentina, March 7-11, 2022. (talk)
- XVI Young Researchers Workshop in Geometry, Mechanics and Control. Barcelona, Spain, December 1-3, 2021. (talk)
- VIII Congress of Applied, Computational, and Industrial Mathematics. La Plata, Argentina, 2021. (talk)
- 15th International Young Researchers Workshop on Geometry, Mechanics and Control. Netherlands, by zoom, November 30 to December 4, 2020. (attended)
- 3rd BYMAT Conference. Valencia, Spain by zoom, December 1-3, 2020. (poster)
- Congress of Argentine Mathematical Union (VirtUMA2020). Argentina by zoom, September 2020. (talk)
- II Argentine meeting of Geometric Mechanics and Mathematical Physics. La Plata, Argentina, June 11-14, 2019. (poster)
- XV Congress Dr. Antonio Monteiro. Bahia Blanca, Argentina, June 5-7, 2019. (talk)

- International Congress of Mathematicians (ICM). Rio de Janeiro, Brazil, August 1-9, 2018. (attended)
- Congress in Honor of Jorge Solomin (SoloFest). La Plata, Argentina, September 15-20, 2018. (talk)
- Congress of woman mathematicians of Latin America. Valdivia, Chile, January 22-26, 2018. (poster)
- LXVI Congress of the Spanish Royal Mathematical Society and Argentine Mathematical Union (RSME-UMA). Buenos Aires, Argentina, December, 2017. (talk)
- I Argentine meeting of Geometric Mechanics and Mathematical Physics. Mar del Plata, Argentina, September, 2017. (talk)
- VI meeting of Science and Technology. La Plata. Argentina, September 27-29, 2016. (poster)
- V meeting of Science and Technology. La Plata. Argentina, November 12-14, 2014. (poster)
- Congress of Argentine Mathematical Union. San Luis, Argentina, September 15-20, 2014. (attended)
- Congress of Argentine Mathematical Union. Rosario, Argentina, September 15-20, 2013. (talk)
- National Congress of students (CoNECEx). La Plata. Argentina, September 6-8, 2013. (attended)

Organization of international conferences in the field of the applicant.

- III Argentine meeting of Geometric Mechanics and Mathematical Physics, La Plata, Argentina, August 24-26, 2022. (Member of the organizing Committee)
- Congress of CMaLP (Centro de Matemática de La Plata, Argentina), La Plata, Argentina, August 22-23. (Member of the organizing Committee)
- Congress of Argentine Mathematical Union. Catamarca, Argentina, July-September, 2024. (Member of the Organizing Committee of the session on Applications of Mathematics and Mathematical Physics)

Research visits, grants, fellowships and awards.

Long term research visits:

- Visiting Professor at the ICMAT, Madrid, Spain, 2021 (3 months).
- Visiting Professor at the ICMAT, Madrid, Spain, 2019 (6 weeks).

Short term research visits:

- Visiting Professor at the National University of the South, Bahia Blanca, Argentina, 2018 (2 weeks).
- Visiting Professor at the ICMAT, Madrid, Spain, 2024 (1 week).

Grants/ Fellowships:

- 2021 – CONICET visiting grant, 3 months, Madrid, Spain.
- 2019 – Severo Ochoa visiting grant, 6 weeks, Madrid, Spain.
- 04/2020 – 06/2023 – CONICET Postdoctoral Fellowship. Argentina.
- 04/2015 – 03/2020 – CONICET Doctoral Fellowship. Argentina.

Participation in founded projects.

I have been a member of the following founded projects.

- Title: Population Control of The Pampas Deer and Greenhouse Gases Data Capture in The Argentinean National Park Campos del Tuyu with Multi-Domain Robots. LINCGLOBAL 2022 (INCGL2022). Principal researcher: Leonardo Colombo. CSIC and Ministry of Science and innovation. Spain. 2022-2024.
- Title: Geometric Mechanics and Mathematical Physics. SX007 (FCE-UNLP). Principal researcher: María Eugenia García and Leandro Martín Salomone. UNLP. Argentina. 2025-2028.
- Title: Geometric Mechanics and Mathematical Physics. 11/X915 (FCE-UNLP). Principal researcher: Marcela Zuccalli. UNLP. Argentina. 2020-2024.
- Title: Geometric Mechanics and Mathematical Physics. 11/X758 (FCE-UNLP). Principal researcher: Marcela Zuccalli. UNLP. Argentina. 2016-2019.
- Title: Geometric Mechanics and Mathematical Physics. 11/X628 (FCE-UNLP). Principal researcher: Marcela Zuccalli. UNLP. Argentina. 2015-2015.
- Title: Mate-Fi to extend to secondary school. University of Engineering (UNLP). Principal researcher: Rossana Di Domenicantonio. UNLP. Argentina. 2019-present.
- Title: New tools for teaching mathematics. (FCE-UNLP). Principal researchers: Marcela Zuccalli and Claudia Ruscitti. UNLP. Argentina. 2018-present.

Scientific, mentoring/advising, management, communication and outreach activities.

Scientific: **Referee** of international journal paper: Journal of Geometry and Physics (2022). **Member** during 2019 in advisory commission of Ordinary Professor in Analysis area from FCE-UNLP (Expte. 700-001312/14-00). **Member** during November 2024 of the advisory committee involved in the internal selection process for full-time Full Professor and Associate Professor (FCE-UNLP). **Referee** of international journal paper: Geometric Mechanics (2025). **Referee** of international journal paper: Mediterranean Journal of Mathematics (2025).

Management, communication and outreach: **Founder** and **member** during 2016 the CMaLP. **Member** during 2020- 2025 in the teaching and organization commission of the Mathematics Department (FCE-UNLP). **Member** during 2020-2021 and 2023-2024 in the Department council of Mathematics Department (FCE-UNLP). **Member** during 2025-present in the finance commission of the Mathematics Department (FCE-UNLP). **Member** during 2025-present of the Mathematics Center of La Plata (CMaLP).